COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	Sep 17, 2026	**DATE SUBMITTED:**	
NAME:	Panti, Dominic Rodnee A.	**INSTRUCTOR:**	Sir Jimlord Quejado

Hands-on Activity 9.1 Tree ADT

1. PROCEDURE OUTPUT

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

```cpp
#include <iostream>
#include "BST.h"

int main(){
    BinarySearchTree<int>* tree=new BinarySearchTree(2);

    std::cout<<"Binary Search Tree"<<std::endl;
    tree->insert(3);
    tree->insert(9);
    tree->insert(18);
    tree->insert(0);
    tree->insert(1);
    tree->insert(4);
    tree->insert(5);

    std::cout<<"pre order output: ";
    tree->preOrder();
    std::cout<<std::endl;

    std::cout<<"in order output: ";
    tree->inOrder();
    std::cout<<std::endl;

    std::cout<<"post order output: ";
    tree->postOrder();
    std::cout<<std::endl;

    return 0;
}
```

```
Binary Search Tree
pre order output: 2 0 1 3 9 4 5 18
in order output: 0 1 2 3 4 5 9 18
post order output: 1 0 5 4 18 9 3 2
```

```cpp
#ifndef BST_H
#define BST_H

template <typename T>
class BinarySearchTree{
    private:
        T key;
        BinarySearchTree<T>* left;
        BinarySearchTree<T>* right;

    public:
    BinarySearchTree(T value);

    void insert(T value);

    void preOrder();

    void inOrder();

    void postOrder();
};

template <typename T>
BinarySearchTree<T>::BinarySearchTree(T value){
    key=value;
    left=nullptr;
    right=nullptr;
}

```

```
30  template <typename T>
31  void BinarySearchTree<T>::insert(T value){
32      if(value<key){
33          if(left==nullptr){
34              left=new BinarySearchTree<T>(value);
35          }else{
36              left->insert(value);
37          }
38      }
39      else if(value>key){
40          if(right==nullptr){
41              right=new BinarySearchTree<T>(value);
42          }else{
43              right->insert(value);
44          }
45      }
46  }
47
48  template <typename T>
49  void BinarySearchTree<T>::preOrder(){
50      std::cout<<key<<" ";
51          if(left!=nullptr){
52              left->preOrder();
53          }if(right!=nullptr){
54              right->preOrder();
55          }
56  }
```

```cpp
template <typename T>
void BinarySearchTree<T>::inOrder(){
    if(left!=nullptr){
        left->inOrder();
    }
    std::cout<<key<<" ";


    if(right!=nullptr){
        right->inOrder();
    }
}

template <typename T>
void BinarySearchTree<T>::postOrder(){
    if(left!=nullptr){
        left->postOrder();
    }

    if(right!=nullptr){
        right->postOrder();
    }

    std::cout<<key<<" ";
}

#endif
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

COMPUTER ENGINEERING

QUEZON CITY

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.